

Product and Category Management System

Abstract

This report presents the design and implementation of a **Product and Category Management System** using a relational database. The system manages product and category information through well-defined tables, maintains relationships using primary and foreign keys, supports CRUD (Create, Read, Update, Delete) operations, and generates category-wise product reports.

Introduction

A Product and Category Management System is an essential component of inventory and retail applications. It enables efficient organization of products into categories, simplifies inventory management, and provides meaningful reports for business analysis. This project demonstrates the implementation of the system using SQL and relational database concepts.

Theory

The system is based on the principles of the **Relational Database Management System (RDBMS)**. Products are linked to categories through **primary key** and **foreign key** relationships, ensuring data integrity and minimizing redundancy. SQL operations are used to manipulate and retrieve product information efficiently.

Objectives

- Design tables for **Product** and **Category** entities.
- Define **primary key** and **foreign key** relationships.
- Store product details such as name, category, price, and stock quantity.
- Perform product insertion, updating, and deletion operations.
- Generate category-wise product reports using SQL queries.

Methodology

1. Create the **Category** table with a unique category identifier.
2. Create the **Product** table with a foreign key referencing the Category table.
3. Insert category and product records into the database.
4. Perform CRUD operations on product data.
5. Execute SQL queries to generate category-wise product reports.

Database Design

The database consists of two main tables:

- **Category:** Stores category ID and category name.
- **Product:** Stores product ID, product name, category ID, price, and stock quantity.

The **CategoryID** acts as the primary key in the Category table and as a foreign key in the Product table to establish the relationship.

SQL Operations

The system supports the following operations:

- Insert new product and category records.
- Update existing product information such as price and stock.
- Delete product records when required.
- Retrieve products based on their respective categories.
- Generate category-wise product reports using SQL queries.

Complexity Analysis

Basic SQL operations such as insertion, updating, and deletion execute efficiently using indexed primary keys. Category-wise report generation involves join operations between Product and Category tables, providing optimized performance for structured datasets.

Results

The implemented system successfully maintains product and category information, enforces relational integrity through primary and foreign keys, performs CRUD operations accurately, and generates category-wise product reports for effective inventory management.

Conclusion

The Product and Category Management System provides a reliable and efficient solution for organizing product information within a relational database. Its structured design ensures data consistency, simplifies inventory management, and supports reporting for business decision-making.

References

1. Oracle Database SQL Language Reference.
2. MySQL Documentation.
3. Microsoft SQL Server Documentation.
4. Elmasri, R., & Navathe, S. B. *Fundamentals of Database Systems*.